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DEVELOPMENT OF A WEB-BASED SENIOR HIGH SCHOOL LEARNING MANAGEMENT SYSTEM FOR DASMARIÑAS INTEGRATED SENIOR HIGH SCHOOL

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**In Partial Fulfillment
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CHAPTER 1 INTRODUCTION

Learning Management Systems (LMS) have become critical tools that assist educators and learners alike by simplifying content dissemination, enhancing communication, and monitoring progress (Furqon et al., 2023). Despite this, most educational institutions continue to be challenged by such issues as disorganized delivery of learning resources, disjointed communication, and minimal accessibility of learning resources. Over time, LMS platforms have evolved to include a variety of features such as gamification. This involves the integration of game-like elements such as points, badges, levels, leaderboards and certificates into the learning process. Gamification transforms traditional learning activities into more engaging and interactive experiences, motivating students to participate actively and consistently (young ,2024). By making the learning journey more enjoyable and competitive in a healthy way, LMS-based gamification has proven effective in increasing student engagement, improving knowledge retention, and fostering a more dynamic educational environment (Tiria et al. 2024).

The project focuses on developing a gamified web-based learning management system (LMS) designed specifically for senior high school students. The goal is to make online learning more engaging and interactive by adding game-like



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features such as points, badges, levels, leaderboards, and certificates. These elements aim to boost student motivation, encourage participation, and make the learning process more enjoyable. Teachers will have their own portal where they can easily upload learning materials, create quizzes, and assign tasks to their students. The system will also allow them to track student progress and give feedback. By combining teaching tools with gamification, the LMS hopes to improve both student performance and the overall online learning experience.

Based on our interview with the client, the current process of delivering learning materials to senior high school students is done manually. Teachers distribute lessons, quizzes, and assignments either by sending them individually through messaging apps like Facebook Messenger or by printing hard copies and handing them out in person. This method, while practical in some cases, can be time-consuming and difficult to manage—especially when handling multiple classes or tracking student submissions. It also lacks a centralized system for organizing materials, monitoring student progress, or providing timely feedback. As a result, both teachers and students face challenges in keeping the learning process efficient, consistent, and engaging.

The motivation for conducting the project entitled Development of a Web-based Senior High School Learning Management System for Dasmariñas Integrated High School comes from the need for a centralized and accessible platform to



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manage learning materials and assessments. Our solution aims to centralize all learning materials, quizzes, and assignments in one accessible platform while incorporating game-like elements to boost student motivation and participation. Through this system, we hope to improve the teaching and learning experience by making it more interactive, efficient, and rewarding for both students and teachers.

Project Context

Based on our interview with our client, Dasmariñas Integrated High School (DIHS), originally established in 1975 as Dasmariñas Relocation Center High School, is a well-known public secondary institution located in Barangay Burol I, Dasmariñas City, Cavite. The school serves a diverse student population of over 10,000 learners, with around 4,300 enrolled in senior high school across Grades 11 and 12. DIHS is supported by a dedicated teaching staff of nearly 400 educators and close to 50 non-teaching personnel, operating within 95 classrooms that facilitate an approximate student-to-teacher ratio of 25:1. The school offers a wide range of academic programs such as Accountancy, Business and Management (ABM), Humanities and Social Sciences (HUMSS), Science, Technology, Engineering and Mathematics (STEM), and various Technical-Vocational-Livelihood (TVL) specializations.

However, the school currently faces challenges because most learning materials are still shared through traditional, manual methods. Teachers usually hand



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out printed worksheets and assignments in class, and updates or reminders are often sent through messaging apps. While this approach works to some extent, it can lead to delays in students receiving materials and makes it difficult to keep track of submitted work. Without a centralized online system to organize lessons, quizzes, and assignments, both teachers and students struggle with managing their tasks efficiently. This lack of a streamlined, interactive platform also means that learning can sometimes feel less engaging and harder to follow.

The main problem at Dasmariñas Integrated Senior High School is that there isn't a single online system where students can easily access all their learning materials. Right now, teachers mostly give out printed worksheets and explain lessons in person, and sometimes they send updates through different chat apps. This can make things confusing because students might get materials late or lose track of assignments. Teachers also find it hard to keep track of who has finished their work or how well students are doing. Plus, without any fun or interactive features, students can sometimes feel bored or less motivated to learn. All of this shows that the school really needs a simple, organized online platform to make learning easier, more exciting, and better for everyone involved.



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Purpose and Description

The Development of a Web-Based Learning Management System (LMS) for Dasmariñas Integrated Senior High School (DISHS) aims to enhance learning accessibility, streamline communication, and improve student engagement. The platform will benefit the following users:

Dasmariñas Integrated Senior High School (DISHS) stands to gain significant advantages from the implementation of Learnify, a digitized Learning Management System (LMS) that modernizes and elevates the educational experience. By centralizing all course materials, assessments, and communications in one accessible platform, Learnify reduces administrative burden and enhances the overall flow of teaching and learning. The integration of gamified features such as points, badges, certificates, and leaderboards boosts student motivation and engagement, making learning more interactive and rewarding. Ultimately, Learnify empowers DISHS to improve academic performance, foster a more engaging learning environment, and support informed decision-making through real-time data and feedback.

DISHS Teachers will benefit from reduced workload, improved classroom organization, and more meaningful student engagement. By enabling real-time performance tracking, direct communication, and simplified lesson delivery, Learnify



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helps teachers save time, stay organized, and focus more on teaching rather than administrative tasks ultimately leading to a more rewarding and productive teaching experience.

Students of DISHS will benefit from a more engaging, supportive, and motivating learning environment through Learnify. The system encourages participation and deeper interest in their studies by making learning more rewarding and interactive. With greater access to course materials anytime, students can take control of their learning pace and improve their understanding of lessons beyond classroom hours. The ability to easily connect with teachers also fosters a stronger support system, helping students feel more confident, guided, and involved in their academic journey.

Researchers will benefit from the field of educational technology, instructional design, and learning management systems (LMS). By examining the implementation and effectiveness of Learnify, a web-based LMS for Dasmariñas Integrated Senior High School (DISHS), this research provides critical insights into how digital platforms can enhance student engagement, accessibility, and learning outcomes.

Future researchers can use the study as a valuable reference, serving as a foundation for their own research. By building upon its findings and methodology, they may find guidance in navigating their studies, ultimately aiding them in achieving



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their objectives. This study can act as a stepping stone, providing essential insights and direction for those seeking to develop and expand similar research efforts.

Objectives of the Study

The general objective of the study is to develop a Web-Based Learning Management System (LMS) called Learnify for Dasmariñas Integrated Senior High School (DISHS) that will help the Super administrator, administrator, teachers, students, and researchers in accessing learning materials and assessments. Specifically, this research aims to assess how Learnify improves accessibility, streamlines communication, enables teachers to track student progress, and facilitates the seamless delivery of quizzes and assignments. Additionally, the study seeks to analyze the platform's impact on student learning outcomes, teacher productivity, and overall classroom interaction.

Specifically, it aims to:

1. Design a system that is capable of:
 - a) Implementing Role-Based Access Control (RBAC) for different users (students, teachers, administrator, super administrator) with defined access permissions.
 - b) Providing a search engine, and
 - c) Providing an assessment module that facilitates the creation, submission, and grading of quizzes and assignments for teachers.



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- d) Ensuring system compatibility with multiple devices, including desktops, tablets, and smartphones.
 - e) Integrating a comments and suggestions section for collaborative learning between students and teachers.
 - f) Supporting user account features (password recovery, profile updates, and secure authentication).
 - g) Providing different dashboards for specific user roles (Super administrator, administrator, teacher, students).
 - h) Implementing a gamification module that encourages student engagement through leaderboards, rewards, and badges.
 - i) Providing a system that supports QR code authentication for secure user login, with a separate QR code used to grant access to learning materials upon successful verification.
2. Develop the system using different software and hardware requirements:
- a) The Learnify system uses various software tools and technologies, including HTML for structuring web pages, CSS for styling, JavaScript for interactivity and functionality, PHP for server-side scripting, XAMPP for local testing, MySQL for database management, VSCode for coding, Bootstrap for responsive design, and Google App Script for automation and webhooks.



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b) The hardware environment includes Windows 11 Operating System on an Intel i7 processor with 8GB RAM.

3. Test the system using unit testing, integration testing, performance testing and system testing.

4. Evaluate the system using ISO 25010 standard; and

5. Prepare an implementation plan for the deployment of the system.

Time and place of study

The study officially began in November 2024 at Dasmariñas Integrated Senior High School (DISHS), located in Dasmariñas, Cavite, Philippines. The school serves as the primary setting for the development and testing of Learnify, the proposed Web-based Learning Management System (LMS). The research took place in the academic year 2024–2025, with various phases including planning, data gathering, system design, development, testing, and evaluation being carried out over several months. Initial data collection, interviews, and consultations with students, teachers, and administrators are conducted within DISHS to ensure the system meets the specific needs of its users. The project's final defense was held in May 2025.



Scope and Limitation

The study focuses on the design and development of a web-based learning management system with gamified features, created specifically for the Senior High School Department of Dasmariñas Integrated High School. The system is intended for use by senior high school students and teachers, aiming to improve the delivery and management of learning materials, quizzes, and assignments through a centralized and user-friendly platform. One of the main goals is to enhance student engagement by integrating game-like elements such as badges, points, leaderboards, and certificates to make learning more interactive and motivating. The system allows teachers to upload learning materials, create quizzes and assignments, and automatically generate grades based on student submissions. It also enables students to access resources, submit their work online, and track their performance. The grading system is limited to quizzes and assignments only.

The proposed system consists of four (4) users: The System Administrator, Administrator, Teacher, Student.

System Administrator is responsible for managing the overall technical operations of the platform. This includes maintaining the system's security, ensuring smooth performance, fixing technical issues, managing backups, and updating the



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platform when needed. The System Administrator also creates and manages user access levels for administrator accounts and ensures that the system runs without interruptions or bugs.

Administrator oversees the general operation and management of the system at the institutional level. Their responsibilities include creating teacher and student accounts, registering tracks and strands, subjects ,sections ,assigning subjects to teachers and generating system usage reports.

Teachers are responsible for uploading and organizing learning materials such as lesson notes, modules, quizzes, and assignments. They are also in charge of checking submissions, grading student work, and providing feedback. They serve as the primary content creators and facilitators of learning within the platform.

Students use the system to access their lessons, download learning materials, take quizzes, submit assignments, and monitor their grades. Their main responsibility is to actively participate in the learning process by completing tasks on time and making full use of the platform's resources for their academic success.

There are (7) modules in the proposed system; User Account Management Module, Report Management Module, Record Management Module, Security and Session Management Module, Gamification Module, Backup and Restore Module , System Configuration & Management Module.



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User Account Management Module this module handles user registration, profile updates, and password recovery. It implements Role-Based Access control (RBAC) to make secure, role-specific access for all users.

Report Management Module this module creates CSV reports on user activity, student progress, and academic structure. It helps Administrators track data like enrollments, students, sections and subjects easily.

Record Management Module this module stores and organizes grades and assignments for teachers and It makes findings and managing records simple for Administrators.

Security and Session Management Module let Super Administrator protect the systems with secure logins, like QR Codes, and it also manages user sessions to stop unauthorized access.

Gamification Module adds fun features like points, badges, leaderboards, and certificates. These make learning exciting and encourage students to join in.

Backup and Restore Module this module saves important data automatically or manually. It helps Super Administrator recover data quickly if something goes wrong.



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System Configuration & Management Module let Super Administrator control/set up the system. This keeps the platform running smoothly and meeting school needs.

However, the application comes with some limitations that users should be aware of. It does not include support for Virtual classrooms which mean students cannot participate in live video sessions with teachers through the platform as integrating such a feature requires additional payment that the researchers could not afford. There are limited third-party integrations, as the application primarily supports spreadsheets via API but lacks full compatibility with tools such as Google Docs, Google Meet, and AI-based grading tools. Limited compatibility with older devices and browsers, being optimized for modern setups like Google Chrome (version 85 and above), Mozilla Firefox (version 60 and above), and Microsoft Edge (Chromium-based). Users with outdated systems, such as Windows 7 or older versions of Android/iOS, or those relying on unsupported browsers, may encounter performance issues, user interface glitches, or restricted access.



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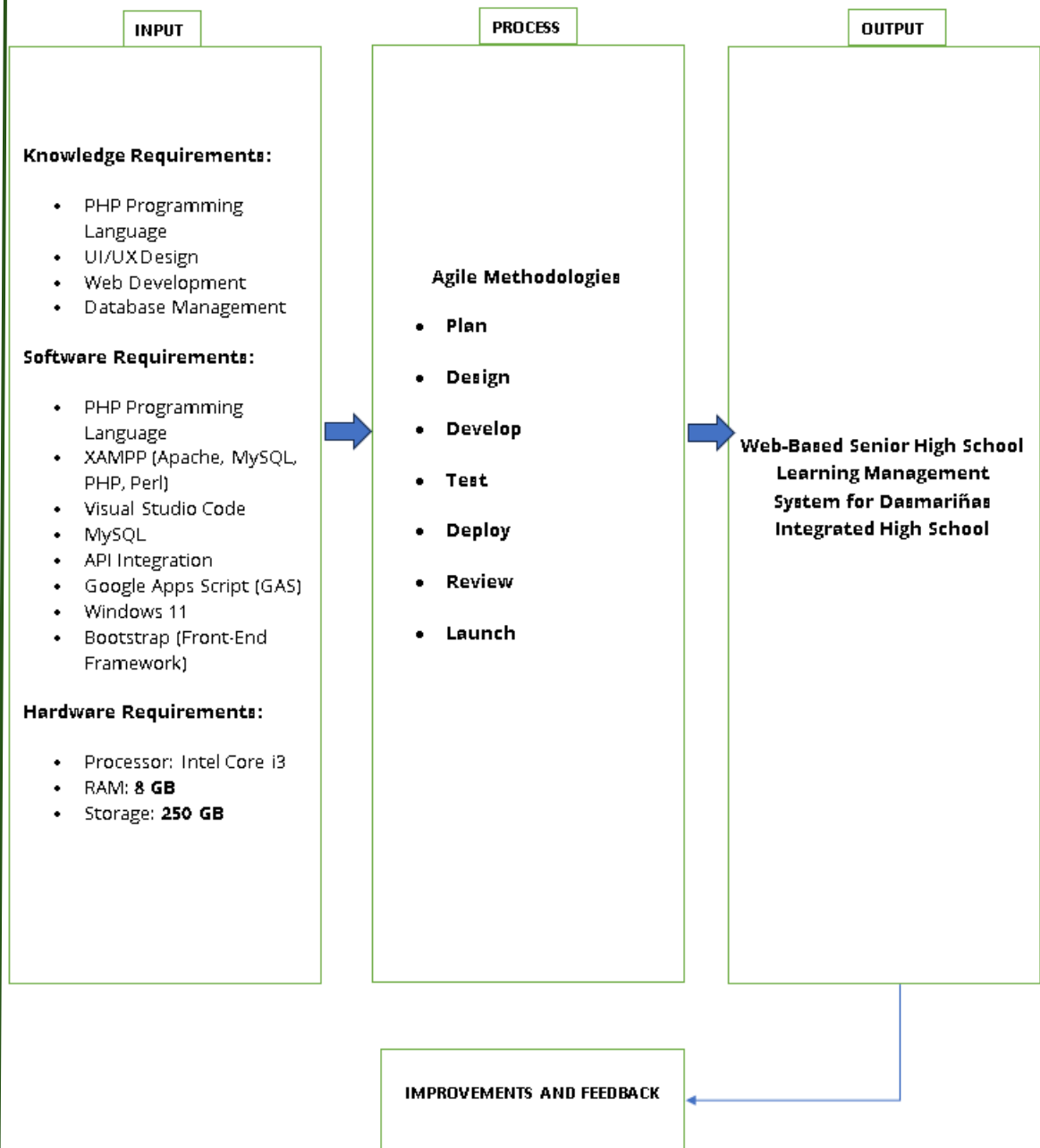


Figure 1. Conceptual Framework of Web-Based Senior High School Learning Management System for Dasmariñas Integrated High School

This conceptual framework illustrates the development process of a Learning Management System for Dasmariñas Integrated Senior High School (DISHS). The framework consists of three main components: Input, Process, and Output (IPO).



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The Input section outlines the necessary knowledge, software, and hardware requirements for the project. Essential knowledge areas include PHP programming, user interface/user experience design, web development, and database management. The software requirements include tools such as XAMPP, Visual Studio Code, MySQL, API integration, Google App Script (GAS), Windows 11, and Bootstrap. Additionally, the hardware requirements specify a minimum processor of Intel Core i3 (8th Gen or higher), 8GB RAM, and 250GB SSD storage.

The Process section follows an Agile Methodology, which includes various development phases: Plan, Design, Develop, Test, Deploy, Review, Launch. In the planning phase the researcher identified the project goals and setting roadmap, in the requirements gathering the researcher defined the system requirements and designing its structures, In the implementation the researcher will code and proceed to system development, In the testing phase the researcher will check for bugs and functionality issues, and will analyze performances and making improvements, and lastly for deployment the researcher will make the system live and accessible. These steps ensure a systematic and thorough process for developing the LMS.

The output of the process is the Development of a Web-based Senior High School Learning Management System for Dasmariñas Integrated High School. This system is designed to access learning materials, provide assessments, and to play fun educational games.



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The model also includes an Evaluation phase to assess the system's performance and ensure it meets the objectives and expectations of its users.

In the evaluation phase, the ISO 25010 is used to measure the quality characteristics of the proposed system. This international standard provides a framework for evaluating software quality across different dimensions. During this phase, the system will be assessed based on relevant criteria such as: Functional Suitability, Reliability, Usability, Performance, Efficiency, Compatibility, Maintainability, and Portability.

General Definitions of Terms

Auto Logout - End user session after inactivity for security.

Class List - Shows students enrolled in a subject or section.

Dashboard - Combination of visual and graphs placed together to understand some analytics and visual key data.

DISHS - Dasmariñas Integrated Senior High School

Gamification - Add points, levels, badges, and certification to boost learning management.

Learnify - Web-based system for managing learning tasks.



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Learning Management System (LMS) - Platform for organizing educational content and progress.

Learner Reference Number (LRN) - Unique 12-digit ID for student login.

Senior High School - Grade 11 and 12 with specialized tracks.

Strand - Specific study area within a track, like STEM.

Track - Broad study category, like Academic or TVL.

Technical Definitions of Terms

API - Tool for fetching user data to enhance user functionality.

Bootstrap - Framework for creating mobile-friendly layouts.

CSS - Styles web interface for a clean look.

Google App Script - Scripting tool for automating data backups.

HTML - Language for building web page structure.

Javascript - Programming language for interactive web features.

MySQL - Database system for storing academic records.

PHP - Programming language for handling backend tasks like user login.

VSCode - Coding tool for developing and debugging

XAMPP - Local server tool for testing web systems.



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Chapter II

REVIEW OF RELATED LITERATURE

Foreign Literature

Learning Management System (LMS) Based On Moodle to Improve Students Learning Activity

The study by Simanullang and Rajagukguk (2020) explores the implementation of a Learning Management System (LMS) based on Moodle to enhance students' learning activities. Moodle, as an open-source LMS, offers a variety of features such as video integration, discussion forums, quizzes, and chat functionalities, which collectively foster an interactive and engaging learning environment. The research highlights the effectiveness of Moodle in addressing challenges like limited face-to-face interaction in traditional classroom settings, thereby promoting continuous and comprehensive learning.

The authors employed a quasi-experimental design to evaluate the impact of Moodle-based LMS on student engagement and activity. Their findings underscore the potential of e-learning platforms to bridge gaps in conventional education systems, particularly in regions with limited access to educational



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resources. This aligns with global trends emphasizing the integration of technology in education to improve accessibility, efficiency, and learner outcomes.

Furthermore, the study contributes to the growing body of literature advocating for the adoption of LMS platforms in educational institutions. It emphasizes the role of technology in transforming traditional pedagogical approaches, making learning more flexible and student-centered. The research also provides insights into the practical applications of Moodle, serving as a valuable reference for educators and policymakers aiming to implement similar systems in their contexts.

Analysis of Development Trends and Experience of using LMS in Modern Education: An overview

The study by Reyad Rajab Aljad (2023) provides an insightful analysis of the development trends and experiences associated with Learning Management Systems (LMS) in modern education. It highlights the transformative role of LMS platforms in enhancing the teaching and learning process, particularly in the context of digital education. The research emphasizes the adaptability of LMS in addressing diverse educational needs, from content delivery to student engagement and performance tracking.



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Aljad's work underscores the importance of integrating LMS into educational frameworks to foster a more personalized and effective learning experience. The study explores various trends, such as the increasing reliance on cloud-based LMS solutions and the incorporation of artificial intelligence to tailor learning paths for individual students. Additionally, it examines the challenges faced by educators and institutions in adopting LMS, including technical barriers and the need for professional development.

This research contributes to the broader discourse on the evolution of educational technology, offering valuable insights for policymakers, educators, and developers. It aligns with global efforts to leverage technology for improving educational outcomes and ensuring accessibility for learners across different contexts.

Active Student Learning through Gamification in a Learning Management System

Learning Management Systems (LMSs) play a key role in online education, especially during situations like the COVID-19 pandemic. However, many traditional LMS platforms face challenges such as keeping students engaged, maintaining classroom efficiency, and preventing distractions. To improve student participation,



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researchers suggest adding game-like features such as points, badges, and leaderboards to make learning more interactive.

A study by Handayani, Raharjo, and Putra (2021) explored how gamification affects online learning. Using a mix of surveys, student discussions, and interviews, they found that students responded positively to features like weekly activity ratings, badges, and leaderboards. Assigning points for different tasks helped boost student motivation and participation in online classes.

These findings suggest that schools and LMS developers should consider adding gamification elements to create a more engaging and effective learning environment. Gamified learning can help solve common issues with traditional online education, making lessons more interactive and enjoyable for students.

Design of web-based LMS (Learning Management System) in SMAN 1 Kampar Kiri Hilir

The study by Muhardi, Gunawan, and Irawan (2020) focuses on the design and implementation of a web-based Learning Management System (LMS) for SMAN 1 Kampar Kiri Hilir. The research highlights the potential of LMS platforms to address challenges such as limited time, space, and interaction opportunities in traditional classroom settings. By leveraging web-based technologies, the study demonstrates



how LMS can serve as a medium for disseminating subject materials, distributing assignments, and facilitating student discussion forums.

The authors utilized the Waterfall Model Method for system development, employing PHP and MySQL as the primary technologies. The resulting LMS application includes features for managing curriculum data, semester schedules, and user roles (admin, teacher, and student). Teachers can upload materials and assignments, while students can access and complete these tasks online. This approach not only enhances teacher-student interaction but also supports continuous learning outside the classroom.

This research contributes to the growing body of literature on the integration of technology in education, emphasizing the importance of user-friendly and accessible LMS platforms. It provides valuable insights for educators and developers aiming to implement similar systems in their institutions, particularly in regions with limited resources.

Learning Management System (LMS) Use with Online Instruction

Learning Management Systems (LMS) are now widely used in education, especially for online learning. Bradley (2021) explains that these systems help teachers organize lessons, communicate with students, and manage coursework.



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LMS platforms make it easier for students to access learning materials, join discussions, and complete assignments in one place.

LMS platforms encourage students to take part in learning by offering interactive features such as discussion boards, quizzes, and multimedia content. Bradley (2021) points out that teachers play a key role in keeping students interested by actively participating in these online platforms. When teachers engage with students through LMS, it helps them stay motivated and focused on their studies.

Although LMS offers many benefits, teachers need to use them properly to be effective. Bradley (2021) stresses that technology should support learning, not replace good teaching methods. Educators must ensure that the LMS is used in a way that aligns with course goals, making learning both engaging and effective.

To make the most of LMS, schools and universities need to provide proper training and research on how these systems impact student learning. Bradley (2021) suggests that this is especially important in subjects like math, where digital tools can help students better understand complex topics. When schools invest in LMS training for teachers, they can improve the quality of education.

In summary, LMS platforms have changed the way students learn by making education more organized and interactive. Bradley (2021) highlights that teachers, schools, and students all have a role in making LMS successful. With the



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right approach, these systems can help improve learning experiences and student engagement.

Local Literature

The Effectiveness of Learning Management Systems in Senior High School Education

Dela Cruz (2021) was a study on the impact of Learning Management Systems (LMS) on Senior High School (SHS) learning in Metro Manila schools. The study was conducted to identify the impacts of Google Classroom and Moodle platforms on the academic performance of students, particularly on the STEM and HUMSS strands. The study concluded that the use of LMS greatly contributed to the positive impact on student cooperation, interaction, and academic performance. The study validated the use of online platforms in the construction of learning spaces to improve interactive and higher learning outcomes.

In the same way, the study in Integrated Senior High School in Dasmariñas quantify the effect of LMS integration on learning experience of the students. Initial findings indicate that LMS integration in teaching increased student engagement, reduced miscommunications between instructors and students, and made learning materials accessible. This was supported by Dela Cruz's (2021) study, validating the application of digital learning technologies in contemporary education.



Amidst the Online Learning Modality: The Usage of Learning Management System (LMS) and Its Relationship to the Academic Performance of the Filipino Students

Tus, J., Cruz, M., Espiritu, N., & Paras, N. (2021) also researched the effect of Learning Management Systems (LMS) on the academic performance of Filipino students when the country transitioned to online learning. Their research was published in the **International Journal of Psychology and Behavioral Sciences**, where it looked into how the inclusion of LMS in the education system has influenced student participation, understanding, and overall performance.

The research highlights that LMS systems provided Filipino learners with easy access to learning materials, interactive software, and instant feedback, which facilitated learning outcomes. The research also, however, noted hindrances such as internet connectivity, lack of adequate technical support, and failure to adjust to web-based learning environments.

This study is relevant to this current study because it presents an overview of how an LMS can be leveraged to improve learning and academic achievement among Filipino learners. It also contributes to the development of a web-based Learning Management System for Integrated Senior High School in Dasmariñas by



emphasizing the importance of user-friendly and well-supported online learning systems.

Students Perceived Satisfaction and Academic Performance Utilizing a Gamified Learning Management System for Senior High School

Cajilla and Bug-os (2022) conducted a study titled "Students' Perceived Satisfaction and Academic Performance Using a Gamified Learning Management System for Senior High School." In this study, published in Science International, the impact of a gamified Learning Management System (LMS) on student satisfaction and academic performance was explored. Results showed that the incorporation of game-like features in an LMS greatly enhanced student engagement, motivation, and learning. The current research is also in line with the goals of the proposed Learning Management System over the web for Integrated Senior High School in Dasmariñas, as the research focuses on the power of technology-based instruction in enhancing learning experiences of the students. Integrating gamification techniques, the project seeks to develop an interesting and interactive learning environment over the web, to enhance the learning outcomes of students.



Gamifying Senior High School Genetics: Students' Experiences and Attitude

Tiria, Misleng, and Caballes (2024) studied the impact of gamification on the learning experience and learning attitude of Senior High School Genetics. Their article, as documented in the International Journal of Scientific Research in Multidisciplinary Studies, presents how game-like features are incorporated into learning materials to induce higher student engagement, motivation, and knowledge retention.

This research is in line with the web-based Learning Management System (LMS) design of Integrated Senior High School in Dasmariñas because gamification is one of the key features of modern digital learning systems. Through the use of interactive quizzes, achievement badges, and progress tracking, the LMS can enhance the interactive nature of the learning process for the students. The findings of Tiria et al. (2024) affirm that gamified learning can improve the performance and attitude of students toward challenging subjects, which makes the potential value of incorporating similar features in the proposed system even stronger.

Impact of the Utilization of Learning Management System in the Implementation of Distance Learning in Secondary Schools in the Philippines from 2019 to 2023

A research project "Impact of the Utilization of Learning Management Systems in the Implementation of Distance Learning in Secondary Schools in the Philippines (2019–2023)" (Journal of Community Development Research, 2024) analyzed how LMS



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websites have impacted online schooling in the nation. The study pointed out that LMS websites improve accessibility and enhance the effectiveness of online learning by offering systematic and organized learning spaces for both learners and instructors. However, the study also identified some of challenges with LMS implementation, including insufficient digital infrastructure, sufficient user training, and a frequent need for technical support. Such challenges are mostly applicable to the creation of a **Web-Based Learning Management System for Integrated Senior High School in Dasmariñas, since they highlight the need to attend to both technology and user issues in using LMS.

Taking into consideration the findings of this local research, the envisioned LMS seeks to provide a better digital learning experience that is responsive to the needs of Senior High School students and teachers. The success factors are a simple interface, efficient content management, and sound technical support, providing a smooth and effective learning environment.



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Foreign Studies

Implementation of Learning Management System: A Way Ahead on the Digital Journey in Distance Learning

The study "Implementation of Learning Management System: A Way Ahead on the Digital Journey in Distance Learning" by Sidra Noreen (2020) explores the integration of Learning Management Systems (LMS) in distance education at the Allama Iqbal Open University (AIOU). The research highlights the transformative impact of LMS on teaching and learning practices, replacing traditional paper-based methods with digital solutions. It emphasizes the advantages of LMS, such as improved accessibility to learning materials, enhanced communication between students and instructors, and the ability to conduct assignments and examinations online.

Additionally, the study employed qualitative methods, including focus group discussions and semi-structured interviews with students and academics, to gather insights into the benefits and challenges of LMS implementation. While the findings reveal numerous advantages, such as increased efficiency and flexibility, they also identify challenges like limited internet access and the need for technical training for both learners and educators. The research recommends establishing small campuses in rural areas and providing mandatory training to address these issues.



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Furthermore, contributes to the digital transformation in education, offering valuable insights for institutions aiming to implement LMS in distance learning contexts. It underscores the importance of addressing infrastructural and training needs to ensure the successful adoption of LMS and enhance the overall learning experience.

Perception of Learning Management System (LMS) on the Academic Performance of Undergraduate Students during the COVID-19 Pandemic

The study by Odekeye, Fakokunde, Metu, and Adewusi (2023) investigates the perception of undergraduate students regarding the use of Learning Management Systems (LMS) in the teaching-learning process during the COVID-19 pandemic. The research highlights the pivotal role of LMS in ensuring the continuity of education amidst global disruptions caused by the pandemic. It emphasizes the advantages of LMS, such as accessibility to learning materials, flexibility in learning schedules, and enhanced communication between students and instructors.

However, study employed a descriptive survey design, collecting data from 551 undergraduate students across five colleges in Osun State University, Nigeria. The findings reveal that LMS significantly contributed to identifying students' knowledge gaps and improving their academic performance. However, challenges such as power outages, poor internet connectivity, and the cost of accessing LMS platforms



were identified as barriers to effective utilization. The research recommends improving infrastructure and providing training to enhance computer literacy among students and educators.

Moreover, study contributes to the broader discourse on the integration of technology in education, particularly in the context of emergency remote learning. It underscores the importance of addressing infrastructural and technical challenges to maximize the benefits of LMS and improve educational outcomes.

Factors that influence online learning assessment and satisfaction: Using Moodle as a Learning Management System

The study by Uthman Alturki and Ahmed Aldraiweesh (2021) examines the application of Learning Management Systems (LMS) during the COVID-19 pandemic, focusing on a sustainable acceptance model through the expansion technology approach. Conducted at King Saud University in Riyadh, Saudi Arabia, the research highlights the critical role of LMS in ensuring the continuity of education during unprecedented global disruptions. It emphasizes the advantages of LMS, such as accessibility to learning materials, flexibility in learning schedules, and enhanced communication between students and instructors.



However the study employs a mixed-methods approach, combining quantitative and qualitative data to assess the effectiveness of LMS in supporting e-learning. Findings reveal that LMS platforms significantly contributed to maintaining academic performance and engagement during the pandemic. However, challenges such as technical barriers, resistance to change, and the need for training were identified as obstacles to effective implementation. The research proposes strategies to address these challenges, including improving infrastructure, fostering user acceptance, and integrating advanced technologies to enhance the LMS experience.

Postgraduate Students' Experiences on the Use of Moodle and Canvas Learning Management System

In their 2020 study, Mpungose and Khoza examined postgraduate students' experiences with Moodle and Canvas Learning Management Systems (LMS) in South African higher education institutions. The research revealed that students primarily utilized these platforms for downloading readings and participating in online discussions, indicating a limited engagement with the broader functionalities of LMS. This suggests a need for more comprehensive training and support to encourage the effective use of all LMS features. [SPRINGER LINK](#)

Similarly, a study by Alharbi and Drew (2021) explored factors influencing postgraduate students' intention to use LMS, identifying performance expectancy,



social influence, and effort expectancy as significant determinants. These findings highlight the importance of user-friendly interfaces and the role of social contexts in promoting LMS adoption.

Furthermore, research by Al-Motham et al. (2022) examined instructors' perceptions of Moodle, considering variables such as gender, teaching experience, and field of study. The study found that these factors significantly influenced instructors' perceived efficacy of Moodle in delivering online education.

Collectively, these studies underscore the importance of comprehensive training, user-friendly design, and consideration of social factors in enhancing the effective use of LMS platforms like Moodle and Canvas in higher education.

The Learning Management System (LMS) and Student Learning Effectiveness: A Systematic Literature Review

Rekha A. P.'s (2024) systematic literature review investigates the effectiveness of Learning Management Systems (LMS) in enhancing student learning outcomes. The study highlights LMS as a powerful tool for planning, implementing, and assessing educational processes. By consolidating evidence from multiple sources, it emphasizes the role of LMS in addressing the challenges of traditional educational methods, such as limited interactivity, fixed schedules, and resource constraints.



The review outlines several benefits of LMS platforms, including their ability to provide flexible learning opportunities, foster collaborative learning environments, and facilitate real-time performance tracking. Furthermore, it identifies key factors contributing to the successful adoption of LMS, such as ease of use, accessibility, and the availability of technical support for educators and students. Challenges like technological barriers, reluctance to adopt new systems, and infrastructure limitations are also discussed, with recommendations for improvement.

This systematic review contributes to the global discourse on technology-enhanced learning, offering valuable insights for educational institutions seeking to optimize LMS implementation. It underscores the importance of designing user-friendly platforms and providing robust support mechanisms to maximize their educational impact.

Local Studies

The Development of a Proposed Learning Management System for Senior High Schools in the Philippines

The study by Delos Santos, Durano, and Hortillosa (2023) focuses on developing a Learning Management System (LMS) for senior high schools in the Philippines, specifically for Grade 11 students. Using a descriptive-survey method, they collected



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data from 3,072 stakeholders, including principals, teachers, and students. The findings revealed several challenges in the current senior high school system, such as large classroom sizes, teachers arriving unprepared, and a lack of engaging lessons. Based on these issues, the researchers designed an LMS tailored to address these concerns and improve the learning experience. They employed an Agile development approach to ensure the system was flexible and met the needs of its users. The study concluded that a well-designed LMS could significantly enhance education by offering more interactive and accessible learning resources, and recommended further evaluation of the system's acceptability, functionality, and scalability for widespread implementation.

Model for web-based learning module in senior high school general chemistry

The increasing integration of technology into education, web-based learning modules (WLM) have become a pivotal tool in enhancing student engagement and academic performance. Origenes (2022) focused on implementing WLM in Senior High School General Chemistry, demonstrating its effectiveness in improving student outcomes. The study utilized a pre-test and post-test approach to assess student performance, finding significant improvement in the post-test results. This suggests that WLMs allow students to learn at their own pace, contributing to a more personalized learning experience. The findings align with previous research indicating that online learning modules can be highly effective in science education,



offering students the flexibility to revisit concepts, engage with interactive materials, and address learning gaps. Furthermore, the study was conducted in response to the challenges posed by the new normal education system, where WLMs provide an essential supplemental resource for students in remote or blended learning environments.

In conclusion, Origenes (2022) underscores the potential of WLM to enhance learning outcomes, particularly in subjects like chemistry, where complex concepts can benefit from interactive and self-paced learning approaches. The study provides evidence supporting the integration of technology into the curriculum to improve student performance, engagement and understanding.

LMS Effectiveness in UM Panabo College: Quipper Efficiency

The study by Dr. Amelie L. Chico (2024) explores the effectiveness of Learning Management Systems (LMS), specifically Quipper, in delivering quality instruction within the hybrid education setup at UM Panabo College. The research highlights the pivotal role of LMS in adapting to the "new normal" brought about by the COVID-19 pandemic, emphasizing its ability to bridge the gap between traditional and online learning environments.

Additionally, study examines the efficiency of Quipper LMS in enhancing student engagement, facilitating seamless communication between educators and learners,



and providing a platform for managing academic tasks such as assignments and assessments. Findings reveal that the LMS significantly contributed to improving academic performance and fostering a more interactive learning experience. However, challenges such as internet connectivity issues and the need for technical training were identified as barriers to optimal utilization.

Furthermore, this research contributes to the growing body of literature on the integration of technology in education, offering valuable insights for institutions aiming to implement LMS platforms effectively. It underscores the importance of addressing infrastructural and technical challenges to maximize the benefits of LMS and improve educational outcomes.

Design and Implementation of i-lnhs: A Learning Management System for Lanang National High School

The integration of technology in education has led to the widespread use of Learning Management Systems (LMS), which are essential tools for delivering and managing educational content (Coates et al., 2005).

1 Aguilar et al. (2021) developed and implemented i-LNHS, a tailored LMS for Lanang National High School, highlighting the importance of customizing such platforms to meet specific institutional needs for improved usability and effectiveness. i-LNHS aimed to overcome limitations like restricted access to physical resources and the need for more interactive learning. By incorporating user-friendly



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interfaces, real-time feedback, and collaborative tools, it sought to enhance both teaching and learning.

The findings of Aguilar et al. (2021) align with previous research emphasizing the role of LMS in boosting accessibility, engagement, and efficiency. As Almarashdeh (2016) noted, successful LMS platforms must engage users and provide seamless functionality. i-LNHS demonstrated its ability to support blended and remote learning, particularly crucial during the COVID-19 pandemic, which accelerated digital learning adoption (Dhawan, 2020). The system also improved student engagement and provided teachers with efficient progress monitoring tools, supporting Khan's (2005) emphasis on personalized and self-paced learning.

However, implementing i-LNHS revealed common challenges, including the need for teacher training, technical support, and infrastructure development. These issues echo Selwyn's (2011) argument that successful technology integration requires robust systems and adequate user support. Aguilar et al. (2021) stressed the importance of continuous evaluation and improvement to maintain the LMS's effectiveness and relevance.

In summary, Aguilar et al. (2021) demonstrate the potential of customized LMS platforms like i-LNHS to enhance high school education. By addressing Lananpin National High School's specific needs, i-LNHS showcased how technology can improve accessibility, engagement, and efficiency. However, successful



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implementation requires careful planning, ongoing support, and addressing the challenges of adopting new technologies.

The impact of Learning Management System (LMS) usage on students.

The integration of Learning Management Systems (LMS) in education has significantly influenced how students engage with learning materials, interact with instructors, and achieve academic outcomes. A 2023 study by Questa Soft titled "The Impact of Learning Management System (LMS) Usage on Students" examines the effects of LMS platforms on student performance, engagement, and overall learning experiences. The study highlights that LMS platforms, when effectively utilized, provide students with greater access to educational resources, facilitate seamless communication, and enable personalized learning pathways. These findings are consistent with earlier research by Coates et al. (2005), which emphasized the role of LMS in enhancing the efficiency and flexibility of educational delivery.

The Questa Soft (2023) study underscores the importance of interactive and user-friendly LMS features, such as discussion forums, quizzes, and progress tracking tools, in fostering student engagement and motivation. For instance, real-time feedback and collaborative tools have been shown to encourage active participation and improve learning outcomes (Almarashdeh, 2016). Additionally, the study notes that LMS platforms support self-paced learning, allowing students to access materials anytime and anywhere, which is particularly beneficial for accommodating diverse



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learning styles and needs (Khan, 2005). This adaptability has become increasingly important in the context of the COVID-19 pandemic, which has accelerated the adoption of online and blended learning models (Dhawan, 2020).

However, the study also identifies challenges associated with LMS usage, such as technical difficulties, the need for digital literacy, and the potential for reduced motivation due to the lack of face-to-face interaction. These challenges align with the findings of Selwyn (2011), who argued that the successful implementation of educational technology requires not only robust systems but also adequate training and support for both students and instructors. The Questa Soft (2023) study emphasizes that the effectiveness of an LMS depends on its alignment with institutional goals, the quality of its design, and the level of support provided to users. In conclusion, the Questa Soft (2023) study highlights the significant potential of LMS platforms to enhance student learning outcomes by improving accessibility, engagement, and personalization. However, it also underscores the importance of addressing challenges such as technical barriers and the need for user training to ensure successful implementation. These insights contribute to the growing body of literature on the role of technology in education, emphasizing the need for continuous evaluation and improvement of LMS platforms to meet the evolving needs of students and educators.



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System Technical Background

According to w3Schools (2025), HTML, or Hyper Text Markup Language, is the standard markup language for creating web pages, using a series of elements to structure content and instruct browsers on how to display it, such as labeling headings, paragraphs or links.

According to GeeksforGeeks (2025), CSS, or Cascading Style Sheets, enhances web page design by managing how HTML elements look, using inline CSS for direct element styling, internal CSS within the HTML file, or external CSS in a separate file for uniform styling across multiple pages.

According to Li (2025), JavaScript is a widely-used programming language that powers user interactions on web applications, backend systems, mobile apps, and even desktop applications, with three types of implementation; Inline, internal, and external scripts, allowing developers to add dynamic functionality to websites.

According to WiseUp (2024), Bootstrap is a front-end framework with a vast set of prebuilt components and a responsive grid system, allowing developers to rapidly build professional, mobile -friendly websites.

According to Simplilearn (2021), PHP, or Hypertext Preprocessor, is an open-source, server-side scripting language embedded in HTML for building dynamic, interactive websites, implemented through inline scripts, internal scripts within the HTML file, or external scripts for consistent functionality across multiple pages.



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According to Adam (2024), Google Apps Script is a JavaScript-based platform within Google Workspace that automates tasks in applications like Google Sheets, allowing users to create custom functions, connect to APIs, and implement scripts inline, internally, or externally for enhanced spreadsheet functionality.

According to Oracle (2025), MySQL, pronounced “My ess-cue-el” (commonly “my sequel”), is the world’s most popular open source relational database management system (RDBMS) that uses Structured Query Language (SQL) to manage and query data stored in tables of rows and columns organized by schemas.

According to Tech Pro (2024), XAMPP is a free and open-source cross-platform web server solution stack package that includes Apache, MariaDB/MySQL, PHP, and Perl, enabling developers to set up a local web server for practicing server-side scripting and managing databases with tools like phpMyAdmin.

According to Jack (2022), an API (Application Programming Interface) is a set of rules and tools that enables different software applications to communicate, allowing data exchange or functionality execution between systems, such as connecting a web application to a database or external service.



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Table 1. Table of Comparison

Title	Author/Year	Features	Similarities
Learning Management System (LMS) Based on Moodle to Improve Students' Learning Activity	Simanullang & Rajagukguk (2020)	Offers features like video integration, forums, quizzes, and chats to enhance student activity and interaction	Gamification elements, Modular content delivery
Active Student Learning through Gamification in an LMS	Handayani et al. (2021)	Incorporates points, badges, and leaderboards to increase motivation and participation	Gamified learning features, Student engagement
Design of Web-Based LMS in SMAN 1 Kampar Kiri Hilir	Muhardi et al. (2020)	Uses PHP & MySQL to manage users, curriculum data, and assignments	Web-based architecture, Role-based access, Modular LMS design
The Effectiveness of LMS in Senior High School Education	Dela Cruz (2021)	Found positive effects of LMS (Moodle/Google Classroom) on cooperation and academic performance	Digital content distribution, Improved communication
Students' Perceived Satisfaction Using a Gamified LMS	Cajilla & Bug-os (2022)	Gamified LMS improved satisfaction, engagement, and performance	Badges, Certificates, Leaderboards



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Impact of LMS Use on Distance Learning in the Philippines	Borabo et al. (2024)	Improved accessibility and engagement; identified infrastructure and training gaps	Usability issues addressed, Importance of support/training
Analysis of Development Trends and Experience of Using LMS in Modern Education	Aljad (2023)	Emphasizes AI-powered learning paths, cloud-based LMS, personalization	Modern LMS trends, Student-centered design
LMS Use with Online Instruction	Bradley (2021)	Promotes interaction via discussion boards, quizzes, and media; teacher engagement crucial	Accessibility, Interactive content, Importance of teacher role
Amidst the Online Learning Modality: LMS Use and Academic Performance	Tus et al. (2021)	LMS supported student access and participation but hindered by tech issues	Challenges in LMS adoption, Digital resource access
Gamifying Senior High School Genetics	Tiria et al. (2024)	Used game mechanics to improve motivation and retention in genetics	Gamification module, Motivation through rewards
Implementation of LMS in Distance Learning	Noreen (2020)	Qualitative analysis of LMS benefits in open universities; improved access,	Accessibility, Infrastructure concerns



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		highlighted rural challenges	
Perception of LMS on Academic Performance During COVID-19	Odekeye et al. (2023)	Showed LMS improved performance but had barriers like cost and internet	Flexibility, Real-time feedback, Accessibility issues
Factors Influencing Online Learning Satisfaction Using Moodle	Alturki & Aldraiweesh (2021)	Mixed-methods study; highlights sustainability and need for user training	Adoption barriers, Need for user training
Postgraduate Experiences on Moodle and Canvas	Mpungose & Khoza (2020)	Limited feature usage due to lack of training; recommends usability improvements	UI/UX importance, Need for training
Systematic Review: LMS and Student Learning Effectiveness	Rekha A.P. (2024)	Reviews global LMS implementations; highlights flexibility, performance tracking	Real-time analytics, Continuous improvement
Development of a Proposed LMS for SHS in the Philippines	Delos Santos et al. (2023)	Agile-based LMS design; responds to lack of engaging lessons and large class sizes	Modular design, Local adaptation, Teacher feedback loop
Web-Based Learning Module for Senior High Chemistry	Origenes (2022)	Uses interactive modules for self-paced learning; improves performance	Supports science learning, Self-paced review



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LMS Effectiveness in UM Panabo (Quipper)	Chico (2024)	Hybrid delivery, improved academic performance; faced tech issues	Quizzes and assignments online, Need for technical support
Design and Implementation of i-LNHS (LMS for Lananpin NHS)	Aguilar et al. (2021)	Customized LMS with interactive tools and real-time feedback	Modular, Context-specific LMS, Emphasis on UI/UX
Impact of LMS on Students (Questa Soft)	Questa Soft (2023)	Focus on interactivity, self-paced learning, real-time feedback	Flexible access, Digital engagement tools

Synthesis

The reviewed local and foreign literature collectively demonstrates the significant role of Learning Management Systems (LMS) in transforming modern education. Foreign studies highlight the impact of platforms like Moodle (Simanullang & Rajagukguk, 2020), the effectiveness of gamification features such as points and leaderboards (Handayani et al., 2021), and the importance of adaptive, cloud-based, and AI-integrated tools (Aljad, 2023) in enhancing student engagement and addressing limitations of traditional learning. These findings emphasize that technology, when effectively integrated, can create more personalized, interactive, and accessible educational experiences. However, challenges such as limited internet



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access, lack of user training, and infrastructural constraints persist, reinforcing the need for institutional readiness and ongoing technical support.

Local studies support these insights and contextualize them within the Philippine education system. Research conducted by Dela Cruz (2021) and Tus et al. (2021) confirms the positive impact of LMS platforms like Google Classroom and Moodle on student engagement, communication, and academic performance in senior high schools. Studies focused on gamified LMS platforms (Cajilla & Bug-os, 2022; Chico, 2024) further validate how integrating game-based learning elements can improve student motivation, satisfaction, and academic outcomes. Moreover, development-focused research (Aguilar et al., 2021; Delos Santos et al., 2023) underscores the effectiveness of customized LMS solutions tailored to the needs of specific institutions.

Together, these sources highlight the potential of LMS platforms to support hybrid and remote learning environments, improve instructional delivery, and facilitate more effective assessment and progress tracking. They also point to the necessity of developing user-friendly, accessible systems with strong technical foundations and responsive support mechanisms. This body of research provides a solid foundation for the development of Learnify—a web-based LMS tailored to Dasmariñas Integrated Senior High School supporting its goal of improving accessibility, student engagement, and academic efficiency through digital tools and gamification.



Chapter III

METHODOLOGY

This chapter discusses how the researchers developed the Web-based Senior High School Learning Management System for Dasmariñas Integrated Senior High School (DISHS). It writes step by step the process that the researchers did to plan, design, construct, and test the system. It also comes with the way the researchers collected information, what instruments they employed, and how they ensured the system functions in a satisfactory manner for both students and teachers. Following this process, the researchers were able to remain focused and accomplish the objectives of the project.



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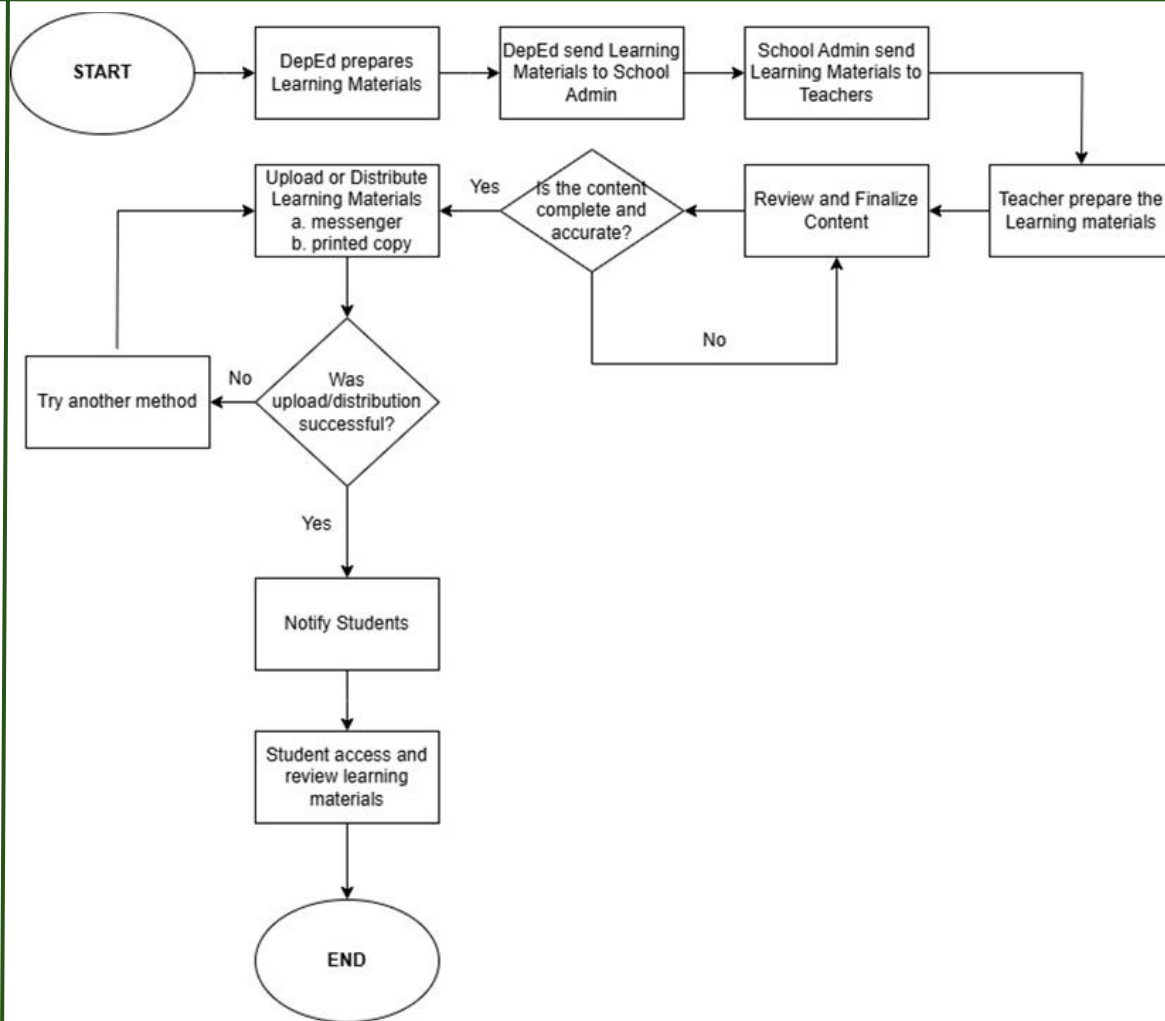


Figure 2. Flowchart of the Current Process for Distributing Learning Materials of Dasmariñas Integrated Senior High School

The flowchart outlines the distribution process of learning materials from the Department of Education (DepEd) down to the students. It starts with DepEd preparing the learning materials, which are then forwarded to the school administration.



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The school admin passes these materials on to the teachers, who are responsible for reviewing, finalizing, or preparing the content. Once the materials are ready, their completeness and accuracy are verified. If the content is found to be incomplete or inaccurate, it goes back for further review and finalization. If the materials are complete and accurate,

The next step involves distributing them to students either through digital means (like Messenger) or physical printed copies. If the distribution or upload is unsuccessful, an alternative method is attempted until the process succeeds. Once the materials are successfully distributed, students are notified. Finally, students access and review the materials, concluding the process.



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Requirement Documentation

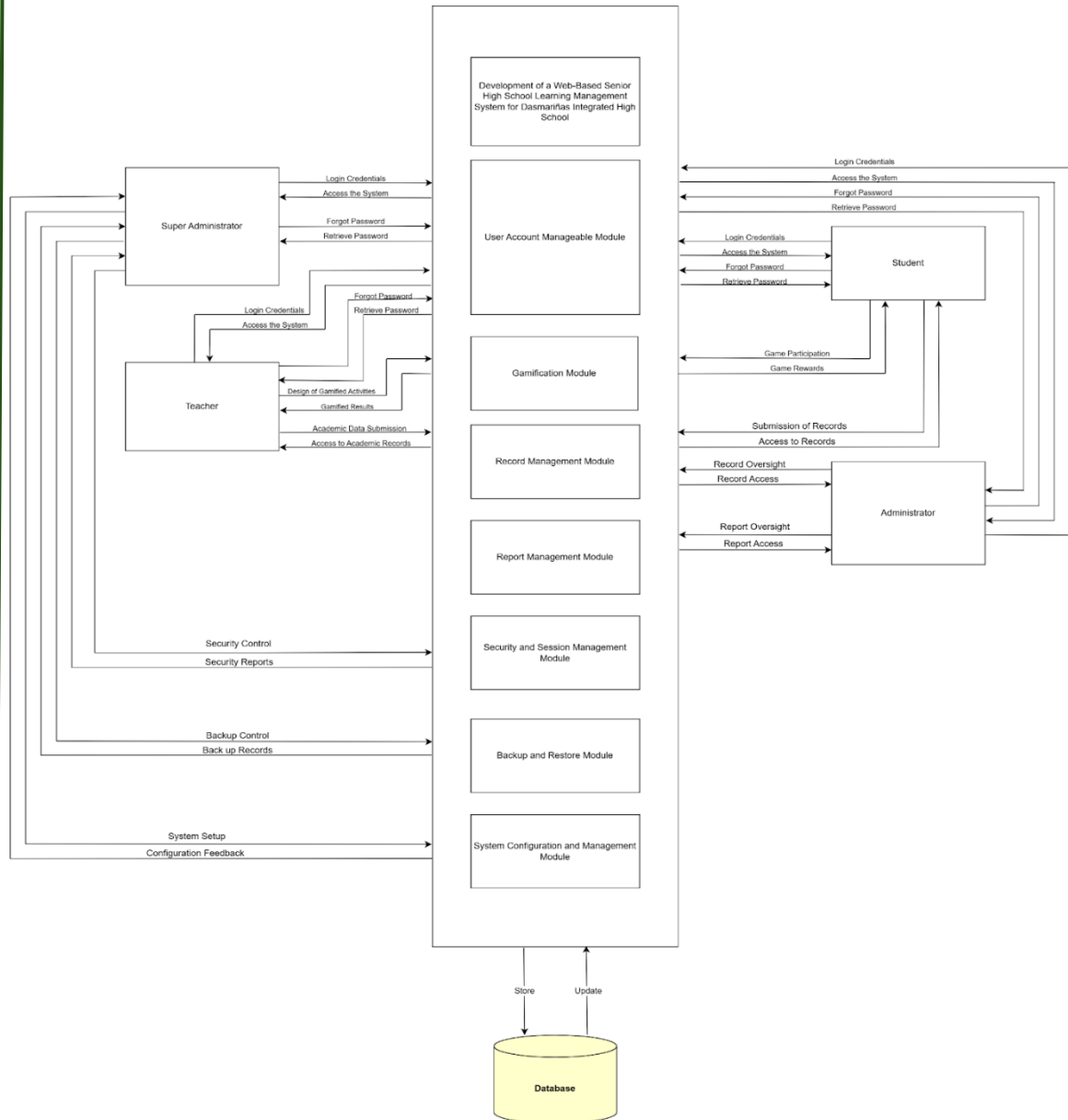


Figure 3. Theoretical Framework for Development of a Web-Based Senior High School Learning Management System for Dasmariñas Integrated High School

The theoretical framework illustrates the structure and interaction of users within a web-based Learning Management System (LMS) designed for a senior high



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school. At the center of the framework is the LMS, which integrates multiple functional modules including user account management, gamification, record and report management, security and session management, backup and restore, and system configuration. These modules facilitate various tasks and processes necessary for effective learning and administration.

The framework involves four main user roles: the super administrator, administrator, teacher, and student. The Super Administrator is responsible for overseeing system operations such as managing user accounts, controlling security settings, performing backups, and configuring the system. The Administrator handles record and report oversight, ensuring proper access to academic data. Teachers upload academic materials, submit academic records, and interact with students' performance data. Students engage with the system through academic assessments (quiz and assignment) and learning tasks, submit their academic outputs, and receive feedback.

Data flows between users and system modules are clearly mapped, showing how users provide inputs such as login credentials or academic submissions and receive outputs like reports, feedback, or access to records. All information is stored and updated in a centralized database, ensuring consistency, security, and ease of access.



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Project Design

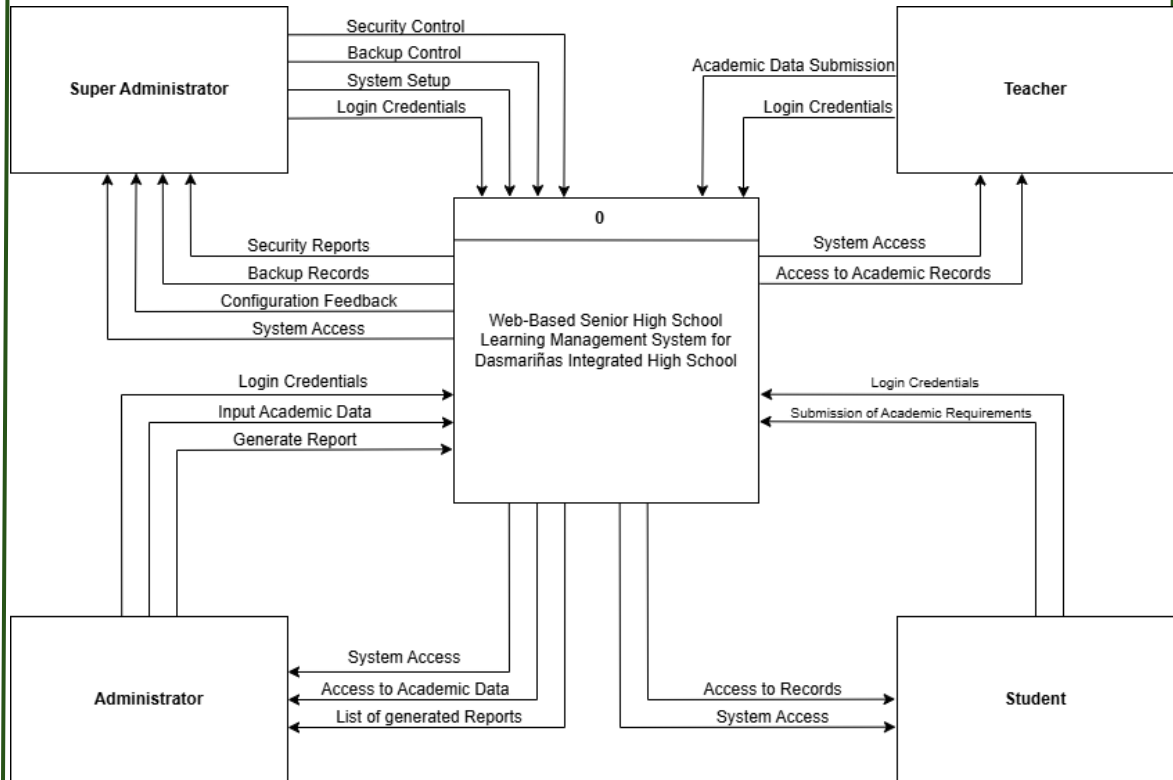


Figure 4. Context Diagram of Development of a Web-Based Senior High School Learning Management System for Dasmariñas Integrated High School

The context diagram represents the overall interaction between the key users and the web-based Senior High School Learning Management System (LMS) for Dasmariñas Integrated High School. At the center of the system is the LMS, which serves as the main platform for managing educational activities and administrative functions. Surrounding it are four main user types: the super administrator,



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administrator, teacher, and student—each interacting with the system in different but interconnected ways.

The super administrator performs high-level management tasks such as setting up the system, managing backups, overseeing security controls, and updating user information. In return, they receive reports and feedback on system performance and security. The administrator accesses and manages academic records and reports submitted through the system. The teacher submits academic data, such as assignments and grades, and interacts with the system to support learning delivery. Students access learning materials, submit their outputs, and receive feedback through the system. This diagram emphasizes the LMS as the central hub that facilitates secure, structured, and role-specific interaction between all users, ensuring efficient educational and administrative processes within the school.



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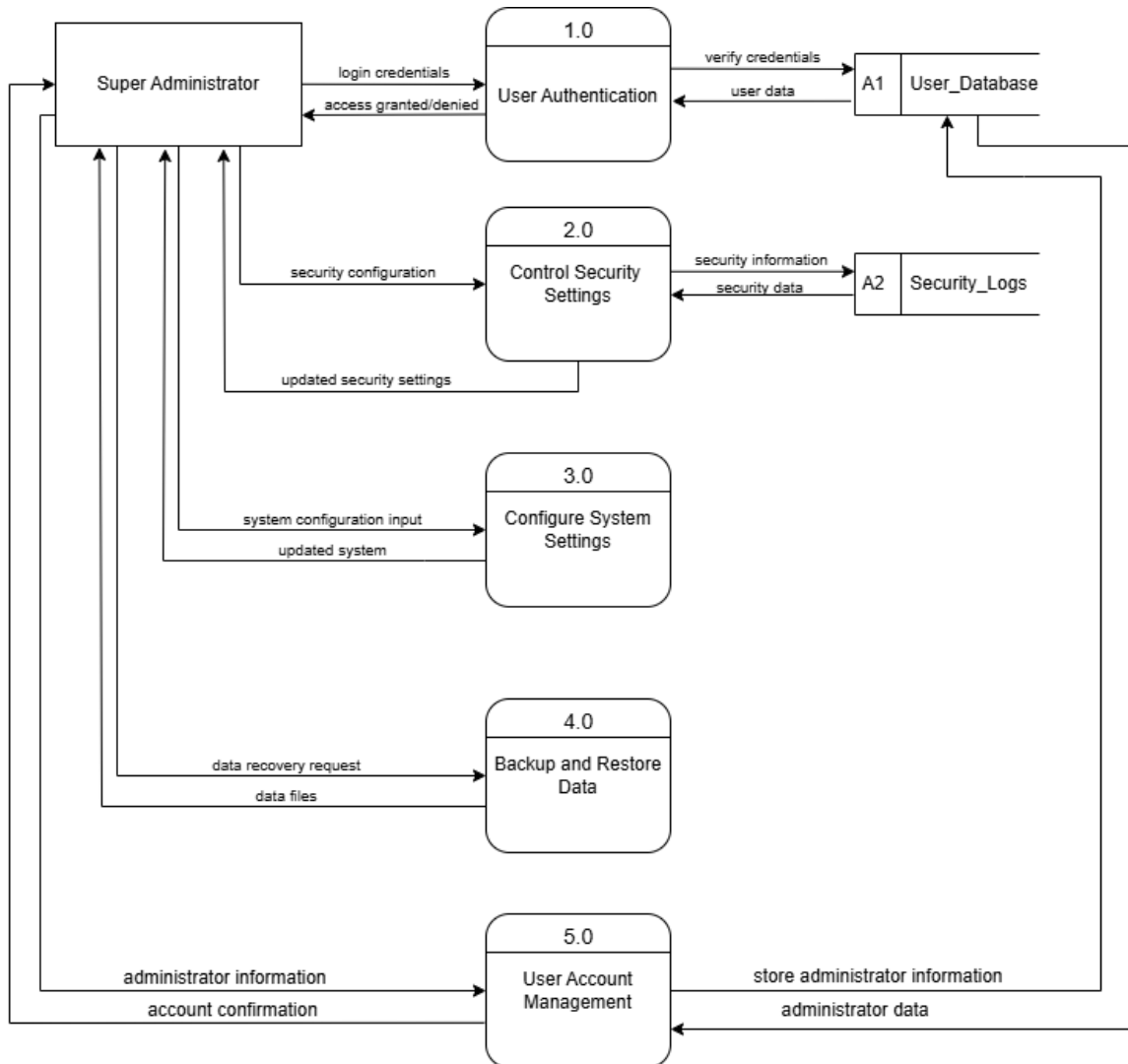


Figure 5. Data Flow Diagram of Super Administrator

This data flow diagram (DFD) illustrates the interactions between the Super Administrator and various core processes in the web-based senior high school learning management system. The Super Administrator first authenticates through the User Authentication process, which verifies login credentials using the User_Database. Upon successful login, the Super Admin can manage system



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operations. They can control security settings—sending configurations and receiving updated logs from the Security_Logs data store. They can also configure system settings, providing inputs and receiving confirmation of updates. Additionally, the Super Admin can initiate data backups or recovery via the Backup and Restore Data process. Lastly, they manage user accounts by sending administrator information to the User Account Management module, which stores the data and returns confirmation.

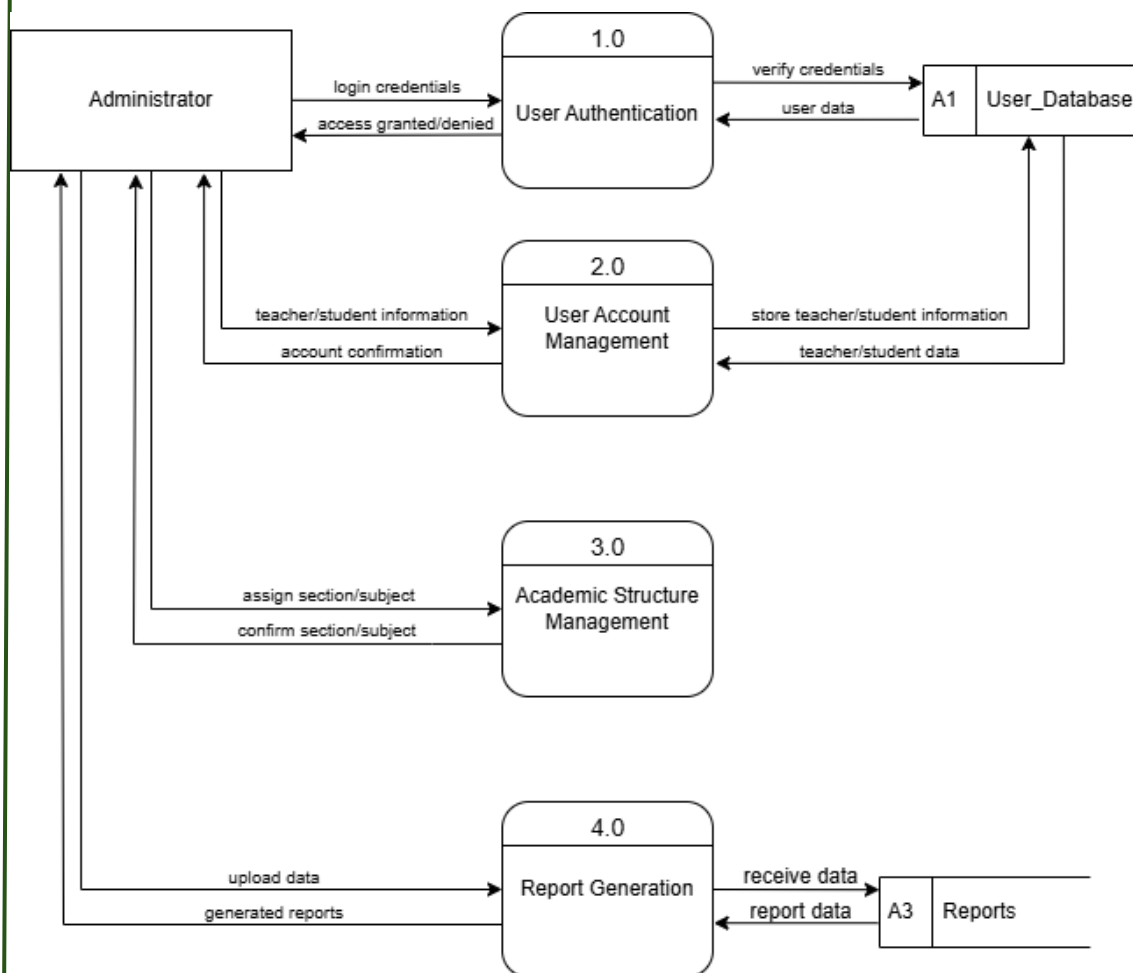


Figure 6. Data Flow Diagram of Administrator



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This data flow diagram (DFD) illustrates the functional role of the Administrator within the web-based senior high school learning management system. The process begins with the Administrator submitting login credentials to the User Authentication module, which verifies them against the User_Database to grant or deny access. Once authenticated, the Administrator can manage user accounts through the User Account Management process by providing teacher or student information, which is stored and retrieved from the User_Database, and receiving confirmation upon successful registration.

The Administrator also handles academic planning through Academic Structure Management, where they can assign sections and subjects to teachers and students. Confirmation is sent back after a successful assignment. Additionally, the Administrator uploads relevant data to the Report Generation module, which processes this information and stores the output in the Reports data store. The Administrator then receives the generated reports for review or distribution.



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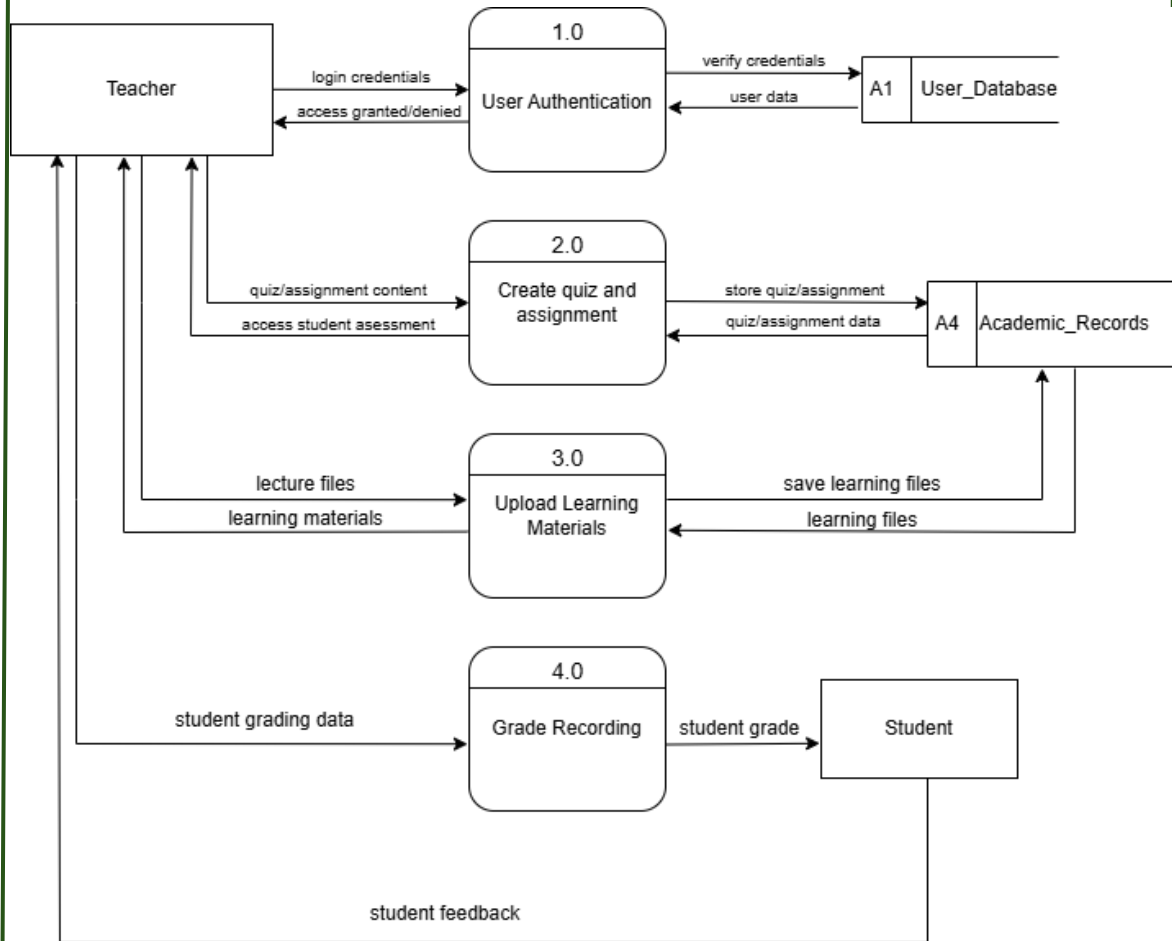


Figure 7. Data Flow Diagram of Teacher

This data flow diagram (DFD) outlines the core responsibilities of a Teacher in the learning management system. It starts with the teacher logging in through the User Authentication process, which verifies credentials using data from the User_Database. Once authenticated, the teacher can proceed to manage academic tasks.



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Through the Create Quiz and Assignment process, the teacher develops assessment content and stores it in the Academic_Records data store. They can also retrieve student assessment data for review. Simultaneously, the Upload Learning Materials module enables teachers to upload lecture files and other instructional content, which is saved as learning files in the Academic_Records.

Lastly, the teacher evaluates students via the Grade Recording process by entering student grading data. The resulting grades are made available to students, who can also provide feedback in return.



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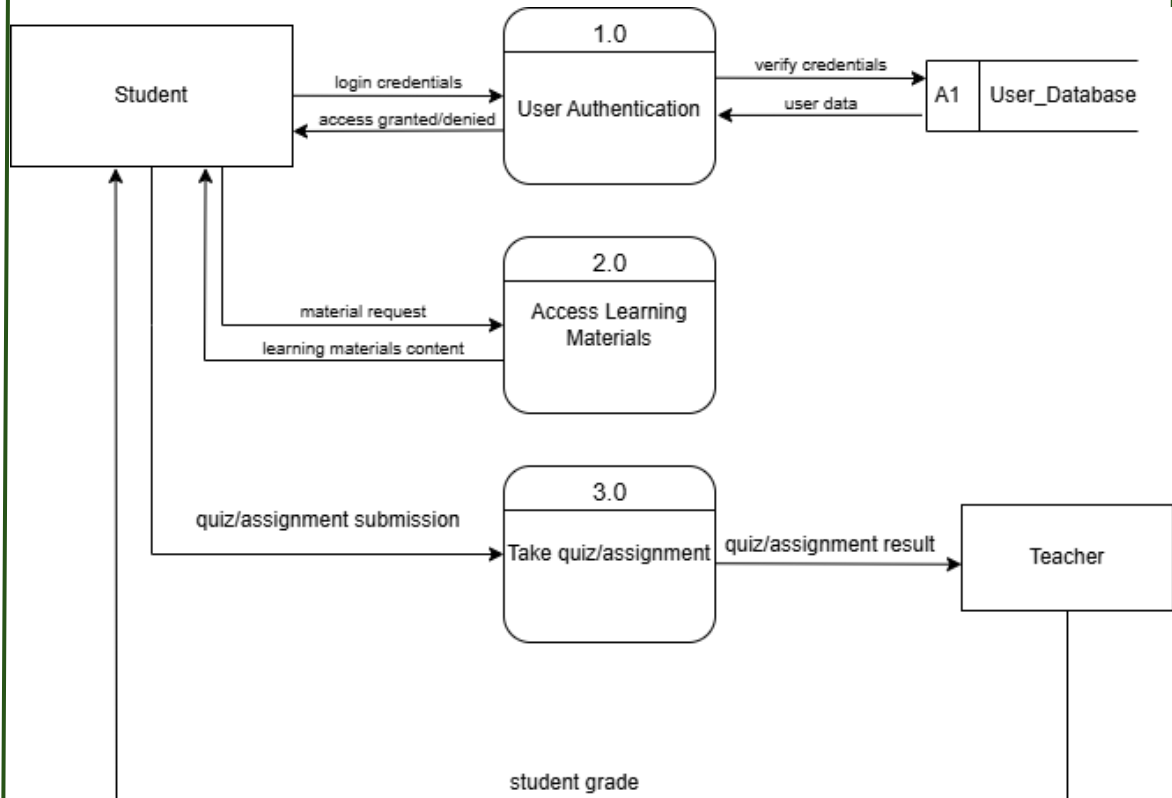


Figure 8. Data Flow Diagram of Student

This data flow diagram (DFD) illustrates the role of the Student in the learning management system. The process begins with the student providing login credentials to the User Authentication system, which verifies their identity through the User_Database. Upon successful login, access is granted.

Once authenticated, the student can access learning materials by sending a material request, after which the system delivers the requested educational content. This supports independent study and review.



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The student can also engage with the Take Quiz/Assignment process, where they submit responses to assessments created by teachers. The teacher receives these submissions and returns the quiz/assignment results. Following this, the student receives their grades as feedback.

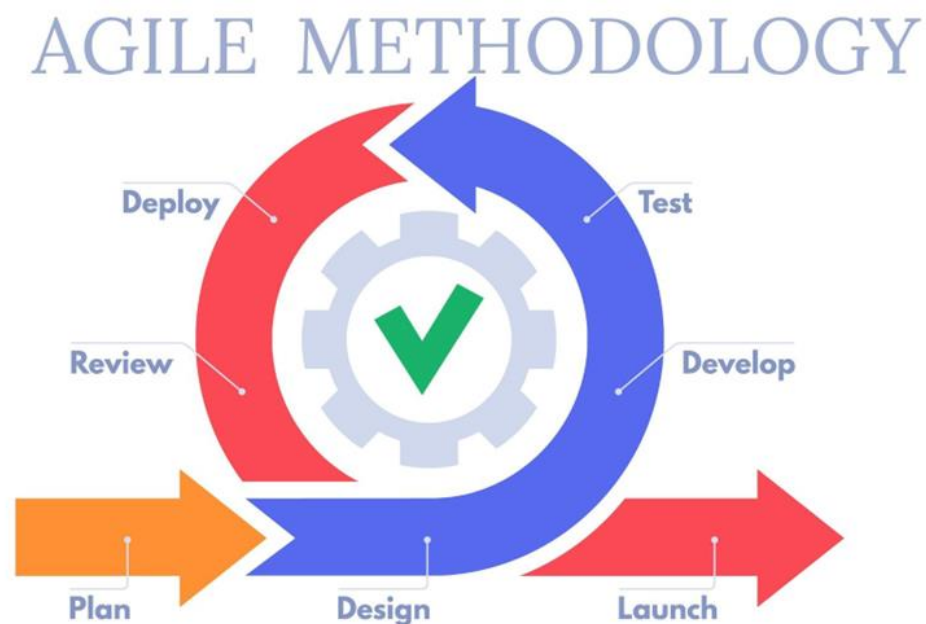


Figure 9. Agile methodology

Planning

During the planning stage, the researchers carried out a requirement gathering to identify the needs and challenges of the DISHS Departments, teachers, and students. This was done through direct interviews and consultation with the DISHS principal, administrators and students, in an effort to understand their current



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approach to managing course materials and administrators works and student engagement and assessment practices.

Design and Development Phase

The Design and Development Phase combines creating and building the features of Learnify step by step in each cycle. Wireframing plays a crucial role in this phase, serving as a visual blueprint that outlines the layout of the platform before actual development begins. These wireframes help streamline decision-making, ensure a user-friendly interface, and guide the team in crafting navigation and interactions. The team works on simple; easy-to-update interfaces and adds features like gamification and QR code login. We use PHP and MySQL for the backend, Google App Script for automation tasks like file handling and email notifications, and HTML, CSS, Bootstrap, and javascript for the frontend.

Testing

During the Testing Phase, the team conducts visual testing, alpha testing, and beta testing within each iteration to ensure system quality. Visual testing verifies the user's interface's design consistency, such as the layout of gamification features. Alpha testing, performed internally, identifies bugs and confirms functionality, like material upload stability. Beta testing involves a limited group of users (administrators, teachers, students) to assess real-world usability, with feedback use to refine the system before the next iteration.



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Deployment

In the Deployment Phase, each iteration's update is deployed to a staging environment on Apache Server for stakeholder testing. A hands-on testing session is conducted with Dasmariñas Integrated Senior High School administrators, teachers, and students to familiarize them with Learnify's features. Such as assessment creation tools and LMS functionalities, enabling early use and feedback. Once these preparations are complete, Learnify is officially launched, making it fully accessible to administrators, teachers, and students.

Review and Feedback Phase

After the system was deployed, feedback was collected from users through direct observation and informal interviews. Admin, teachers and students were encouraged to use the system and share their experiences. Their suggestions and insights helped the development identify areas for improvement. Based on this feedback, minor bugs were fixed, and adjustments were made to enhance the system's overall usability and performance. This phase ensured that the system was refined according to actual user needs and expectations.



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Retrospective Phase

In the Retrospective Phase the process for each iteration is improved to make future work better. Discussions point out what went well, like solid visual testing, and note areas to improve. This phase keeps refining Learnify's Development, following Agile's focus on steady improvement. The original diagram (Figure 9: Agile Methodology) mentions a "Launch" phase, but this process uses an iterative approach with a Retrospective phase, focusing on going deployment and improvement. Learnify becomes usable after the Deployment Phase instead of waiting for one big launch.

Testing Procedure/Tools

Unit Testing. In this phase, each individual module such as user registration, login authentication, file uploading, quiz creation, and gamification is tested in isolation by the development team. The goal is to verify that each module functions correctly according to the system's design. This testing is typically performed by developers during the development phase to identify and fix bugs at the code level before integration with other parts of the system.

Integration Testing. Once the individual modules are confirmed to be working, integration testing is conducted to ensure seamless data flow between modules. This includes verifying that the Student module can access learning



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materials uploaded by Teachers, or that Administrator-generated registration to students data is accurately working. It also ensures that the Super Administrators actions such as managing system settings and restoring backups are accurately reflected across the Learnify. It checks the integration of components like user dashboards and the messaging feature used by both students and teachers.

Performance Testing. The Learnify system is evaluated under expected Dasmariñas Integrated Senior High School (DISHS) usage conditions, particularly during peak periods such as quiz deadlines or mass login attempts. Tools like Google PageSpeed Insights are used to assess loading speeds, while simulated loads help test system stability and responsiveness across desktops, tablets, and smartphones. Performance benchmarks focus on minimizing downtime and ensuring responsiveness of dashboards, quizzes, and the messaging feature.

System Testing. After successful integration and performance evaluations, full system testing is carried out. This involves running the Learnify platform in a controlled environment with actual users (Students, Teachers, Administrator, Super Administrator) from Dasmariñas Integrated Senior High School. The testing process ensures that all user roles can access their respective dashboards, perform their assigned tasks such as uploading, submitting assignments, and experience within a stable and functional system. System testing verifies that the platform aligns with the



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functional requirements set in the objectives and adheres to ISO 25010 quality standards.

Evaluation Procedure

The evaluation of the Learnify Learning Management System will align with ISO 25010 standards to assess its quality and effectiveness.

User Testing and Survey process will be conducted involving students, teachers, and administrator of Dasmariñas Integrated Senior High School (DISHS) to evaluate the Learnify. Teachers, administrator, Super administrator and approximately 200 students of DISHS will engage with the platform, exploring its features and functionalities while providing valuable feedback on their experience. A structured survey will be implemented to gather insights into Learnify functional suitability, usability, reliability, performance, compatibility, security and maintainability based on the ISO 25010 standards, allowing users to rate various aspects of the system. This evaluation process ensures that Learnify meets user expectations and identifies areas for improvement, helping refine the platform for optimal performance in an educational setting. Evaluations will be conducted by numerical and descriptive Scale, ensuring a valid and representative assessment of the system's quality.



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Key criteria include Functional Suitability, Usability, Reliability, Performance, Security, Compatibility, and Maintainability.

Functionality Suitability. Evaluates the Learnify ability to meet user needs by performing the required tasks accurately and effectively. This includes features like module management, gamification elements (e.g., badges, leaderboards), and progress tracking. Additionally, User testing will focus on the effectiveness of functionalities such as uploading assignments, private messages of teachers and students, dashboard changing realtime, creating quiz of teachers and accessing course materials of the students.

Usability. Measures the system's ease of use, ensuring that the interface is intuitive and accessible for students, teachers, and administrators. It is evaluated by its users(students, teachers, administrator, super administrator) based on how effectively they can navigate and interact with the system. A well-structured layout, clear labeling, and properly positioned buttons contribute to a smooth and smooth user experience. Each module should feature thoughtfully placed action buttons to minimize confusion and enhance workflow, whether the task involves uploading materials, creating quizzes, or performing other module-specific actions of Learnify . User feedback will be collected through surveys and interviews to assess the interface's clarity, layout, and overall navigational design.



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Reliability. Assesses the system's ability to perform consistently under specific conditions. This includes availability during school hours and resilience against crashes or errors. Stress testing will simulate high user loads to verify system stability, especially during peak usage.

Performance. Evaluates the system's response times, resource usage, and scalability under different conditions. Tools such as Google PageSpeed Insights will be used to test loading times and system responsiveness of Learnify.

Security. Tools such as Security Headers will help protect user data by setting rules for how the system and browser communicate. These headers will prevent unauthorized access, secure login credentials, and keep stored data safe. They will also help ensure smooth and protected interactions between users and the system, making the Learnify more reliable for administrator, students and teachers.

Compatibility. Checks the system's ability to work seamlessly across different devices, browsers, and operating systems. Tests will confirm compatibility with modern browsers like Google Chrome (version 65 above), Mozilla Firefox (version 60 above), and Microsoft Edge on various devices.

Maintainability. Measures how easily the system can be updated, debugged, and maintained over time. Evaluation of the modular design to ensure quick resolution of technical issues and integration of new features.



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Table 2. System Evaluation Sheet Numerical and Descriptive Scale.

NUMERICAL SCALE	INTERPRETATION	Definition
4.21 – 5.00	Excellent	The system fully meets and far exceeds the most expectations.
3.41 – 4.20	Very Good	The system fully meets all exceeds several expectations.
2.61 – 3.40	Good	The system fully meets all expectations.
1.81 – 2.60	Fair	The system does not fully meet all expectations.
1.00 – 1.80	Poor	The system fails to meet expectations to significant degree in several areas.



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Table 3. Project Implementation Plan

Phase	Activities	Responsible Party	Timeline
1. System Setup	Configure the Learnify platform, set up servers, provide login credentials.	Super Admin	Week 1
2. User Registration	Perform manual registration for Super Admins and Admins: Use bulk upload for teachers and students.	-Super Administrator (for admins) -Administrators for (Teachers / Students)	Week 2
3. Data Configuration	Create and configure Tracks & Strands, Subjects, and Sections; assign Teachers to Subjects and advisory roles.	Administrator	Week 3 - Week 4
4. Content Upload	Upload initial learning materials (modules) and create Word Search and Learnify Quiz games aligned with subjects.	Teachers	Week 5 - Week 6
5. Student Enrollment	Enroll Students into Sections and Subjects; notify Students of their enrollment.	Administrators, Teachers	Week 7
6. Testing and Feedback	Test system functionality (dashboards, CRUD operations, games, messaging); collect feedback from Teachers and Students.	Super Administrator, Administrator, Teachers, Students	Week 8 - Week 9



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7.Revisions & Backup	- apply revisions based on feedback - perform a full database backup for disaster recovery.	Super Administrator	Week 10
8.Final Deployment	Approve final configurations; enable access for all users; activate ongoing features like games and feedback/ratings.	Super Administrator, Administrator	Week 11
9. Monitoring & Support	- monitor logins -ensure notifications - feedback	Super Administrator, Administrator	Ongoing



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Appendix 1. Interview questions

1. What are the current issues or problems your institution faces in managing learning?
2. How do you currently deliver lessons and track student progress?
3. How many students are currently enrolled in your senior high school?
4. How many teachers are assigned to senior high school classes?
5. What challenges do teachers and students experience with the existing institution process?
6. Are there any delays or difficulties in grading and giving feedback?
7. Do students and teachers have trouble accessing learning materials?



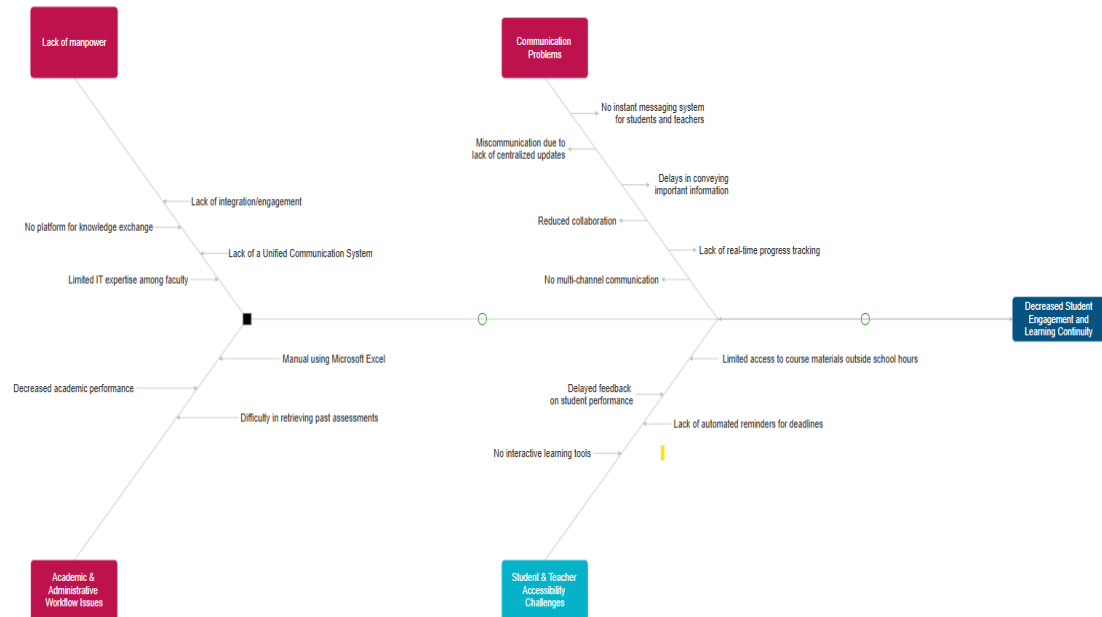
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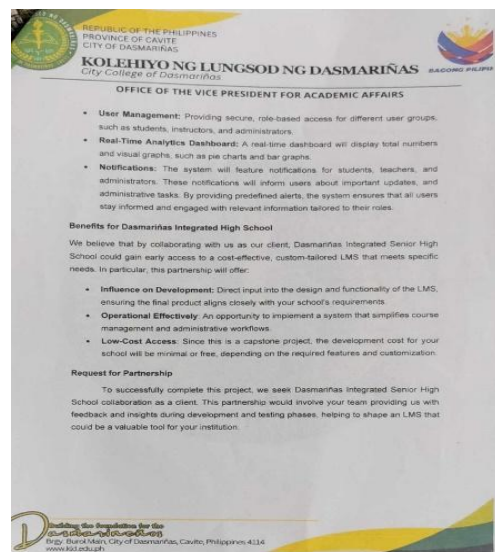
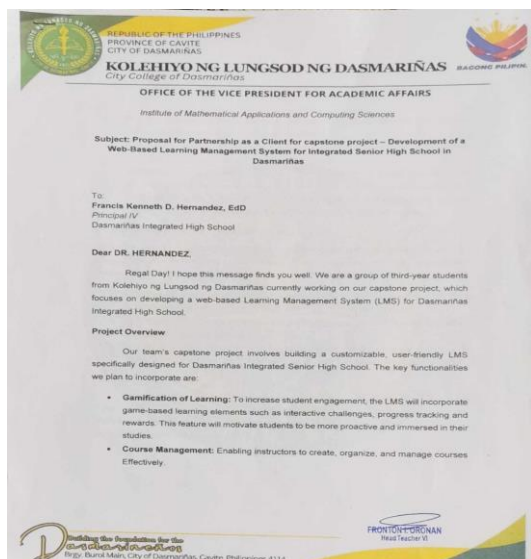
8. What features would make learning more effective for your institution?

9. How do you handle communication between students and teachers?

Fishbone Diagram



Appendix 3. Letter Interview/Proposal





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